

AI for Compliance

BISfit

BIS Standards Recommendation Engine
for Indian Micro & Small Enterprises



BIS STANDARDS
MSME RAG System

Team : Tech Majdoors

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01 | Problem Statement

The MSE Compliance Gap

- > 63 million MSEs in India face complex BIS compliance requirements
- > Weeks spent manually identifying applicable standards
- > 500+ BIS standards in Building Materials alone
- > Non-compliance risks: penalties, market rejection, safety failures
- > No affordable, accessible tool exists for smaller enterprises

6–8 Weeks

Average time to identify applicable BIS standards

63M+

MSEs in India needing compliance guidance

₹2L–5L

Estimated cost of non-compliance per incident

BISfit — Ultra-Fast Standards Intelligence Solution

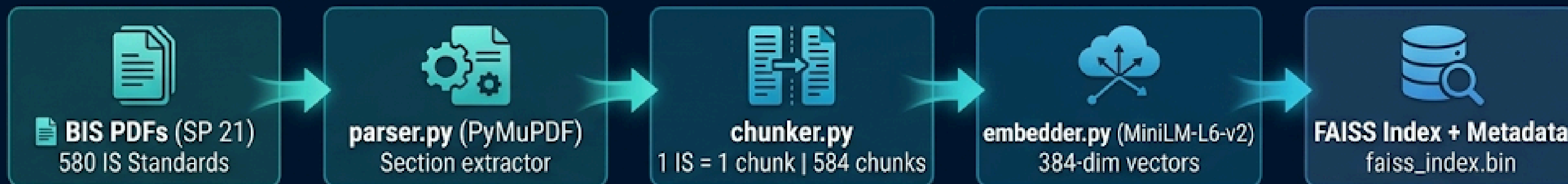
Natural Language Compliance in Seconds, Powered by Optimized RAG



Independent research tool. Not affiliated with BIS.

03 | System Architecture

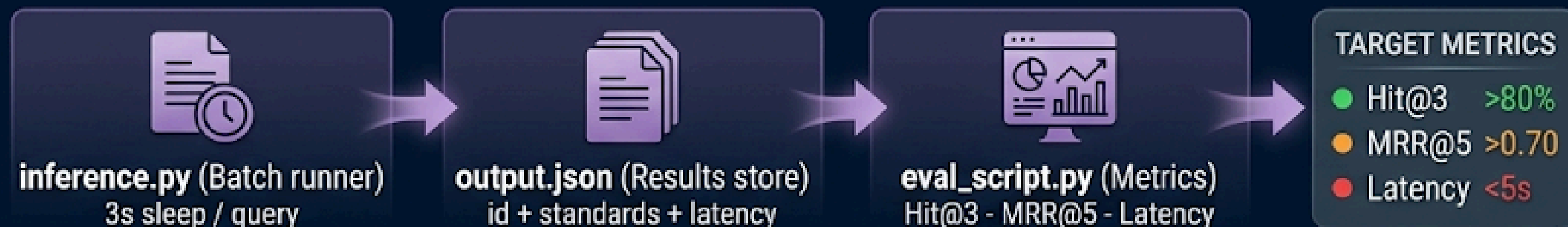
PHASE 1 — OFFLINE INGESTION



PHASE 2 — ONLINE INFERENCE (Per Query)



EVALUATION



Chunking & Retrieval Strategy

584 Chunks · 580 IS Codes

4-Stage Ingestion Pipeline



580 IS Standards

584 Total Chunks

3 Standards Split

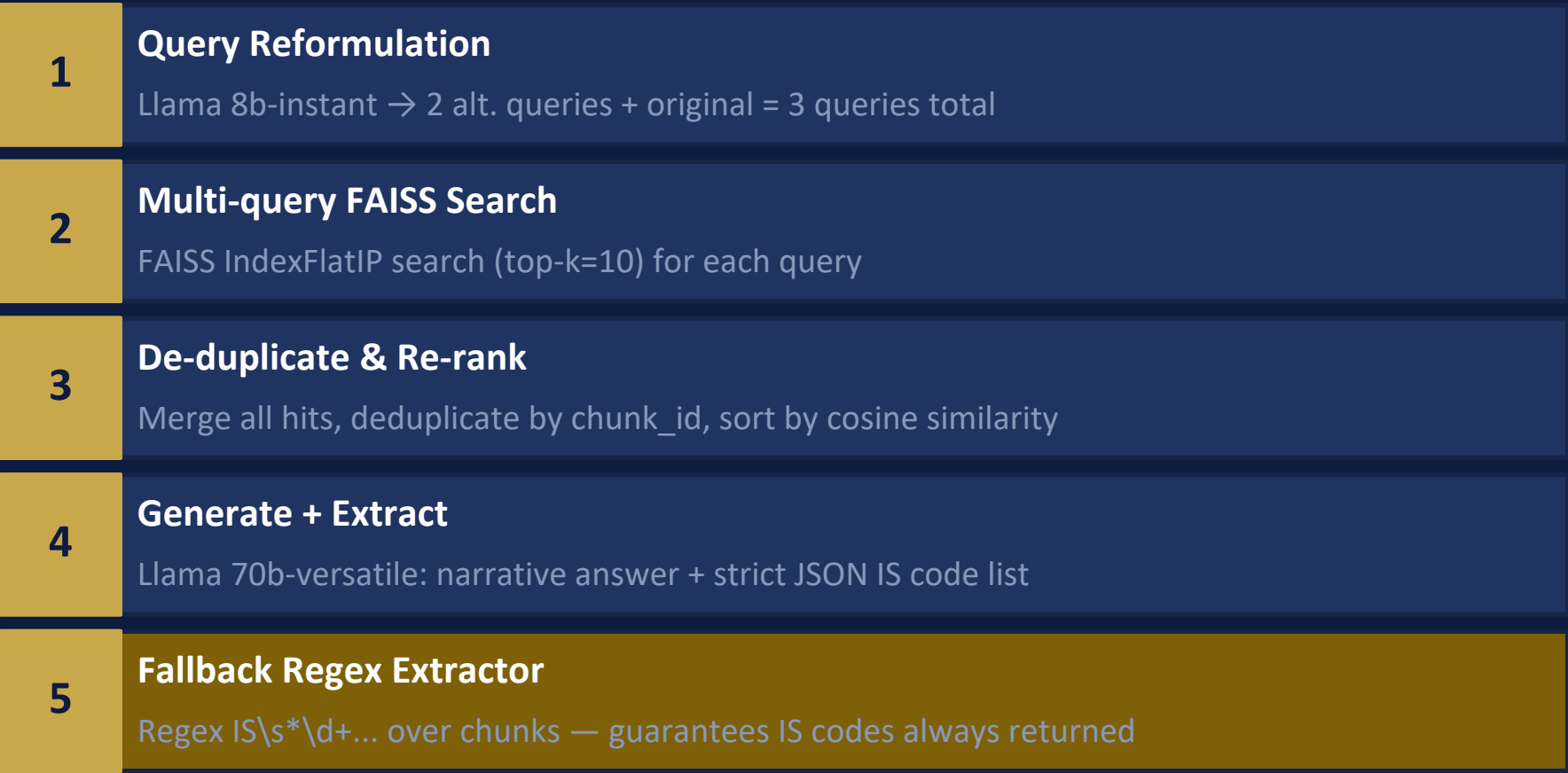
Key Design Decisions

Dual-Field Chunk Design

text_to_embed — Metadata header + content → fed to MiniLM encoder
5-line header: IS Number · Title · Section · Source

content_only — Scope + Requirements only → fed to LLM generation
Saves ~180 tokens of LLM context per query (6 chunks × 30 tokens)

Online Retrieval Flow



384-dim Embeddings

IndexFlatIP · Exact Search

~877 KB FAISS Index

05

| Demo Highlights

Illustrative Example — Cement Manufacturer

User Input	Reformulated Queries
Product: OPC 53 Grade Cement Use: Structural concrete, RCC Role: Manufacturer, BIS license pending	Q1: "OPC 53 grade cement compressive strength IS specifications" Q2: "Portland cement BIS license CRS mandatory requirements" Q3: "cement fineness setting time soundness test IS 269"

Top Recommendations Returned:

IS 269:2015	IS 1489 Pt.1	IS 4031
Ordinary Portland Cement Mandatory: Yes (CRS) Confidence: High → Apply via BIS Simplified Procedure	Portland Pozzolana Cement Mandatory: Yes (CRS) Confidence: Medium → Apply via BIS Simplified Procedure	Methods of Physical Tests Mandatory: Voluntary Confidence: High → Apply via BIS Simplified Procedure

* Placeholder output for illustrative purposes

Evaluation Results

BIS Hackathon — Public Dataset

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Queries Evaluated



Hit Rate @3

90%

Target >80%

✓ EXCEEDED

9 of 10 queries returned
a correct IS code in top-3



MRR @5

0.90

Target >0.70

✓ EXCEEDED

Mean Reciprocal Rank
across top-5 retrieved results



Avg Latency

1.39s

Target <5 sec

✓ EXCEEDED

End-to-end wall-clock time
per query (incl. LLM calls)

07 | Impact on MSEs

From weeks of confusion to minutes of clarity.

BEFORE BISfit

- ❌ 6–8 weeks searching BIS website manually
- ❌ Consultant fees of ₹50,000–₹2,00,000
- ❌ Risk of applying wrong / outdated standard
- ❌ No clarity on mandatory vs. voluntary
- ❌ No actionable next step for certification



AFTER BISfit

- ✅ Standard recommendations within few seconds
- ✅ Zero consultant dependency for initial scoping
- ✅ Grounded strictly in official BIS documents
- ✅ Mandatory / CRS flag shown per recommendation
- ✅ Direct certification pathway mapped per standard

BISfit bridges the compliance knowledge gap for 63 million MSEs — making BIS standards accessible, actionable, and fast.

08 | Team & Acknowledgements

BISfit

BIS Standards Recommendation Engine



Shreyash Golhani

Applied AI and Core ML



Vaageesh Kumar Singh

Agentic AI and Product Mindset

Acknowledgements

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